

Java File Handling and Pattern Counting

Writing text to a file and counting “pe” and “pi” occurrences

Objective / Aim

To write a Java program that stores a passage in a text file, reads the file line by line, and counts the occurrences of the character pairs **pe** and **pi**, without regard to letter case.

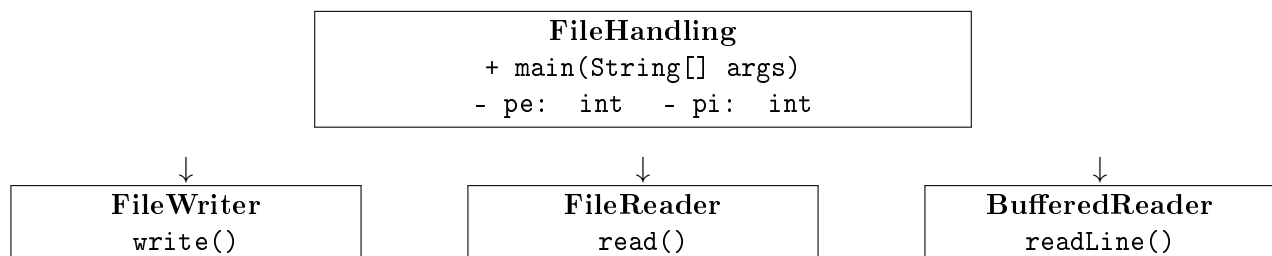
System Requirements

- **Hardware:** Standard computer with a Core i3/i5-class processor and 4 GB or more RAM.
- **Software:** Windows, Linux, or macOS; JDK 17 or JDK 21; and a text editor or Java IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans.

Algorithm / Logic Description

1. Store the four-line passage in a string, using newline characters between lines.
2. Open `sample.txt` with a `FileWriter`, write the passage, and close the writer.
3. Create a `BufferedReader` around a `FileReader` for `sample.txt`.
4. Read each line and convert it to lowercase so that the search is case-insensitive.
5. Examine every adjacent pair of characters in that line. Increment **pe** when the pair is **pe**, and increment **pi** when it is **pi**.
6. Close the reader and print both occurrence counts. If a file operation fails, print the I/O error message.

UML Class Diagram



The program writes the passage with `FileWriter`. It reads the file with `FileReader`, wrapped in a `BufferedReader`.

Source Code

Save the following program as `FileHandling.java`:

```
import java.io.*;
import java.util.*;

public class FileHandling {
```

```

public static void main(String[] args) {

    String text =
        "Peter Piper picked a peck of pickled peppers\n" +
        "A peck of pickled peppers Peter Piper picked\n" +
        "If Peter Piper picked a peck of pickled peppers\n" +
        "Where's the peck of pickled peppers Peter Piper picked?";

    try {

        FileWriter fw = new FileWriter("sample.txt");
        fw.write(text);
        fw.close();

        BufferedReader br =
            new BufferedReader(new FileReader("sample.txt"));

        int pe = 0;
        int pi = 0;
        String line;

        while ((line = br.readLine()) != null) {

            line = line.toLowerCase();

            for (int i = 0; i < line.length() - 1; i++) {

                String pattern = line.substring(i, i + 2);

                if (pattern.equals("pe"))
                    pe++;

                if (pattern.equals("pi"))
                    pi++;

            }

        }

        br.close();

        System.out.println("'pe' - no of occurrences - " + pe);
        System.out.println("'pi' - no of occurrences - " + pi);

    } catch (IOException e) {
        System.out.println("File error: " + e.getMessage());
    }

}

```

Compilation & Execution Commands

Open a terminal in the folder containing `FileHandling.java`, then run:

```

javac FileHandling.java
java FileHandling

```

The program creates `sample.txt` in the current working folder.

Output Screen

The program prints:

```
'pe' - no of occurrences - 20  
'pi' - no of occurrences - 12
```